

Disproof of Conjecture 00000007623

earthking11

6 October 2026

The conjecture says that the nilpotency index is at most two, with a “degenerate” exception of index three. Consider the linear operator on $\mathbb{Q}^4$ whose matrix in the standard basis is

$$J = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

For a coordinate vector, $J(a, b, c, d) = (b, c, d, 0)$. Consequently

$$J^4 = 0, \quad J^3(0, 0, 0, 1) = (1, 0, 0, 0) \neq 0.$$

Thus J is nilpotent of index exactly four. This is neither an index at most two nor the stated index-three exception, so the conjecture is false.

The Lean project encodes the same four-coordinate operator over the integers. The fourth-iterate identity is definitional, while the displayed nonzero third iterate is kernel-checked. No search or unformalized computation is needed.